

Project Summary

CopernicusAI Knowledge Engine: Assisted Research via AI Briefings, Multi-Modal Metadata, and Process Visualization

Principal Investigator: Gary Welz

Program: CISE Core Programs - Information and Intelligent Systems (IIS)

Requested Amount: \$691,184 (3 years)

Overview

This project develops and studies an AI-enabled “assisted research” platform that helps researchers and advanced students rapidly orient to topics and connect knowledge across scientific domains using traceable sources and structured metadata (Welz, 2026a). The core idea is to treat AI-generated briefings—especially interactive audio—as a research interface rather than static science communication. The platform extends RAG with structured analysis, connection discovery, cross-modal integration, and provenance. It links text, audio, video, and diagrams through a unified metadata layer. The work builds on public prototypes: CopernicusAI, Programming Framework, Genome Logic Modeling Project (GLMP) (biological processes as interactive flowcharts), Science Video Database, and Research Paper Metadata Database. A fully operational Knowledge Engine (December 2025) has knowledge graph visualization, vector search, and RAG capabilities.

Intellectual Merit

The project advances human-centered AI-assisted research workflows grounded in multi-source evidence. Contributions: (1) Research briefings as interactive interface: Design and evaluation of “prompt → briefing → refinement” loops for progressive question narrowing. (2) Unified metadata across modalities: Structured JSON capturing provenance, entities, methods, claims, and cross-references for papers, videos, briefings, diagrams, and audio. (3) Reliable AI via multi-checker workflows and collaborative validation: Multiple AI and human checkers, provenance enforcement, citation verification. (4) Process visualization as computable artifacts: Programming Framework converts textual process descriptions into Mermaid diagrams (JSON), demonstrated in GLMP. These align with the Knowledge Engine evaluation agenda (retrieval quality, connection validity, user-reported value).

Broader Impacts

The platform provides a scalable system for research orientation and exploration. The existing system was developed with limited resources, demonstrating that Knowledge Engine–style systems are feasible for individuals and small teams; NSF support enables scaling, evaluation, and broader access through open tools and student involvement. Users generate and share research briefings, enabling collective learning grounded in scientific sources. The system promotes responsible AI through provenance, validation, and multi-checker review. Student users participate in alpha and beta testing. The project includes accessibility (WCAG 2.1 AA) and DEI considerations. All research components will be open-sourced (MIT License) and disseminated through publications and repositories.